

APQP — IATF 16949 CLAUSE MAPPING

Maps all APQP deliverables to IATF 16949 clauses | Audit Readiness Reference | Quality Head Agents

PURPOSE

This document maps all APQP deliverables and activities to specific IATF 16949:2016 clauses. Use this during internal audits, external IATF audits, and APQP gate reviews to demonstrate compliance.

COMPLETE APQP — IATF 16949 CLAUSE MAPPING TABLE

APQP DELIVERABLE	IATF CLAUSE	CLAUSE TITLE	PHASE	OBJECTIVE EVIDENCE REQUIRED
Voice of Customer (VOC)	8.2.2, 8.2.3	Requirements — customer, statutory, regulatory	1	VOC document, customer requirements matrix
Design Goals	8.3.2	Design & development inputs	1	Design goals document signed by customer
Preliminary BOM	8.3.3, 8.4.2.4	D&D controls; Supplier controls	1	BOM Rev 01, supplier PPAP status
Special Characteristics (SC)	8.3.3.1	Product design outputs — supplemental	1	SC register, drawing SC symbols
Management Support Sign-off	5.1.1	Leadership and commitment	1	Signed management support form
APQP Timing Plan	8.1	Planning of product & service provision	1	Timing plan with target/actual dates
DFMEA	8.3.3, 8.3.5	D&D controls; D&D outputs	2	DFMEA report, RPN analysis, action items
Design Verification Plan (DVP)	8.3.4	Design & development review	2	DVP&R document, test results
Engineering Drawings	8.3.5	Design & development outputs	2	Drawing set with SC symbols, approved
Design Review	8.3.4	Design & development review	2	Design review minutes, attendance, sign-off
Team Feasibility Commitment	8.3.2	Design & development inputs	2	Feasibility form signed by all CFT members
Process Flow Diagram (PFD)	8.5.1.1	Control plan	3	PFD with process step numbers matching PFMEA
PFMEA	8.5.1.1	Control plan	3	PFMEA linked to PFD, RPN < threshold, SC addressed
Pre-Launch Control Plan	8.5.1.1	Control plan	3	Control plan signed by quality manager
Process Instructions / WI	8.5.1.2	Standardized work and operator instructions	3	WI at workstations, operator acknowledgment
MSA Plan	7.1.5.1	Measurement resources — general	3	MSA plan listing all critical gauges
Packaging Specifications	8.5.4	Preservation	3	Packaging specification approved
Poka-Yoke Validation	8.3.4.4	Design & development controls	3	Mistake-proofing log, validation records
Production Trial Run	8.5.1	Control of production & service provision	4	PTR report — minimum 300 pcs, SPC data
MSA / GRR Study	7.1.5.1, 7.1.5.2	Measurement resources	4	GRR study results — %GRR < 10% for SC
Process Capability Study	8.5.1.1, 9.1.1	Control plan; Monitoring, measurement, analysis	4	Analysis — Cpk ≥ 1.67 for SC
Production Control Plan	8.5.1.1	Control plan	4	Final control plan signed and issued
PPAP Submission	8.6.1, 8.6.5	Release of products and services	4	Full PPAP package, 18 elements complete
PSW — Customer Approval	8.6.1	Release of products and services	4	Customer-signed PSW, PPAP level confirmed
SPC Implementation	9.1.1.2	Statistical tools	5	SPC charts running, Western Electric rules applied
Lessons Learned	10.3	Continual improvement	5	Lessons learned database entry, future program updates
APQP Closure Review	9.1.3	Analysis and evaluation	5	Closure review report, all issues closed

KEY AUDIT QUESTIONS — IATF AUDITOR PERSPECTIVE

- Q1. How does the organization ensure APQP is applied to all new programs and significant engineering changes?
- Q2. Show me the APQP timing plan and demonstrate that gate reviews were conducted at each phase.
- Q3. How are special characteristics identified in the customer drawing cascaded to PFMEA and Control Plan?
- Q4. Demonstrate that PFD, PFMEA, and Control Plan have matching process step numbers.
- Q5. Show MSA results for all special characteristic measurement systems. What is your GRR%?
- Q6. What is the preliminary Cpk for the safety characteristic on this part? Show objective evidence.
- Q7. How are lessons learned from previous APQP programs incorporated into new launches?